

Stream Cipher Classification Using AI: A Neural Network-Based Approach

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Abstract—This research explores a deep learning-based approach for stream cipher classification using artificial neural networks (ANNs) and convolutional neural networks (CNNs). We generated and labeled keystream datasets from three widely known stream ciphers—RC4, TRIVIUM, and ESPRESSO—at four different keystream lengths: 8, 24, 32, and 64 bytes. For each configuration, a total of 2^{17} samples were generated. Despite experimenting with different neural network architectures and varying keystream lengths, all models consistently yielded an average classification accuracy of approximately 30%. This suggests that the models struggled to distinguish between the ciphers based solely on keystream patterns, raising questions about the feasibility of AI-based classification in this context without deeper cryptographic features. These findings highlight limitations in neural network-based classification of stream cipher outputs and motivate further research into feature engineering and model design for cryptanalysis.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Stream ciphers are lightweight encryption algorithms widely used in real-time and resource-constrained systems due to their high speed and low computational overhead. Unlike block ciphers, stream ciphers encrypt data one bit or byte at a time, making them suitable for applications such as mobile communications, IoT devices, and embedded systems. Popular stream ciphers include RC4, TRIVIUM, and ESPRESSO, each with different internal architectures and statistical properties.

With the rise of machine learning (ML) and deep learning (DL), researchers have started exploring whether neural networks can learn the statistical patterns or structural biases present in the output of cryptographic algorithms. This has sparked interest in applying AI techniques to problems in cryptanalysis, cipher detection, and security assessment.

In this study, we investigate whether deep learning models—specifically artificial neural networks (ANNs) and convolutional neural networks (CNNs)—can classify the keystream outputs of different stream ciphers based solely on raw binary data. We generated a large dataset consisting of 2^{17} keystream samples for each of the three ciphers (RC4, TRIVIUM, and ESPRESSO), across four keystream lengths: 8, 24, 32, and 64 bytes.

Despite varying the architecture and input length, our experimental results consistently yielded an average classification accuracy of approximately 30%—close to random guessing

for three classes. This suggests that the neural networks were unable to learn distinguishable features from the raw keystreams alone. Our findings raise important questions about the limitations of using neural models for stream cipher classification and highlight the need for further research into feature engineering, hybrid approaches, or alternative model designs.

II. RELATED WORK

The intersection of machine learning and cryptanalysis has been increasingly explored in recent years. Several studies have aimed to use statistical techniques or AI models to identify or analyze cipher outputs. Traditional methods often relied on handcrafted features such as entropy, frequency analysis, and correlation patterns to distinguish between encrypted and non-encrypted data [1]. However, such approaches require significant domain expertise and often fail to generalize across cipher families.

More recently, deep learning methods have been applied to cryptographic analysis tasks. For instance, convolutional neural networks (CNNs) have been explored for classifying ciphertexts and detecting anomalies in encrypted traffic [2]. Some works also proposed the use of artificial neural networks (ANNs) for cipher recognition based on learned representations of ciphertext or keystreams [3]. However, these efforts largely focus on block ciphers or high-level network encryption protocols.

To date, very little research has specifically targeted the classification of stream ciphers using deep learning techniques. Moreover, the effectiveness of neural models in distinguishing between cipher keystreams purely based on raw binary patterns remains an open question. Our work contributes to this space by systematically evaluating the classification capabilities of ANNs and CNNs on keystreams generated by three prominent stream ciphers—RC4, TRIVIUM, and ESPRESSO—at varying byte lengths. Noteworthy prior work includes Xiao et al. [?] who proposed neural cryptanalysis metrics using ANN models on proprietary stream ciphers [?], and Akhilesh et al. [?] who applied LSTM networks to the Espresso stream cipher, revealing the potential of recurrent architectures in this area. Another study by Payal et al. [?] investigated keystream analysis of the Grain cipher using CNNs and RNNs. Additionally, Mishra et al. [?] performed deep learning analysis specifically

on RC4 and related ciphers, demonstrating both feasibility and challenges of AI-driven cryptanalysis.

III. PROPOSED METHODOLOGY

In this work, we explore the feasibility of using deep learning techniques—specifically artificial neural networks (ANNs) and convolutional neural networks (CNNs)—to classify the output of stream ciphers. Our methodology consists of three primary stages: keystream data generation, preprocessing, and model design and training.

A. Keystream Dataset Generation

We generated keystream samples from three widely studied stream ciphers: RC4, TRIVIUM, and ESPRESSO. For each cipher, we produced a total of 2^{17} samples at four different keystream lengths: 8 bytes, 24 bytes, 32 bytes, and 64 bytes. Each sample was generated using randomly initialized keys or IVs, ensuring diversity across the dataset. The generated binary outputs were saved in CSV format and labeled according to their corresponding cipher class.

B. Data Preprocessing

The raw keystream outputs were transformed into a flat numerical format. Each byte was converted into its corresponding integer value (ranging from 0 to 255), forming input vectors of fixed length depending on the keystream size (e.g., 64 integers for 64-byte inputs). Labels were one-hot encoded for multi-class classification. The dataset was then split into training and testing sets in an 80:20 ratio and normalized where necessary.

C. Model Architecture

We designed and trained both ANN and CNN models to evaluate the classification performance. The ANN consisted of fully connected layers with ReLU activation and dropout for regularization. The CNN model included one-dimensional convolutional layers to capture local patterns in the keystream data, followed by pooling and dense layers. Both models concluded with a softmax output layer for 3-class classification (RC4, TRIVIUM, ESPRESSO). The models were trained using the Adam optimizer and categorical cross-entropy loss function.

IV. EXPERIMENTS AND RESULTS

To evaluate the performance of our models, we conducted a series of experiments using both artificial neural networks (ANNs) and convolutional neural networks (CNNs) on the generated stream cipher keystream datasets. Each model was trained separately on keystreams of varying lengths: 8, 24, 32, and 64 bytes.

A. Training Setup

The datasets were divided into training and testing sets with an 80:20 ratio. All input data were normalized to ensure consistent learning behavior. For both models, we used the Adam optimizer with categorical cross-entropy loss. The ANN model consisted of multiple dense layers with ReLU activation and dropout for regularization. The CNN model used 1D

convolutional layers followed by pooling and dense layers, also ending with a softmax layer for 3-class classification.

B. Results

Across all experiments, the models consistently achieved an average classification accuracy of approximately 30%, regardless of the input byte size or model architecture. This result is equivalent to random guessing in a 3-class classification task, indicating that the models failed to learn distinguishable features from the cipher keystreams.

This suggests that the keystreams produced by RC4, TRIVIUM, and ESPRESSO do not contain easily learnable statistical differences detectable by standard deep learning models. Table I summarizes the classification accuracy obtained.

TABLE I
CLASSIFICATION ACCURACY (%) ACROSS DIFFERENT BYTE SIZES

Byte Size	ANN Accuracy	CNN Accuracy
8 bytes	30.2	30.1
24 bytes	29.8	30.3
32 bytes	30.0	30.0
64 bytes	30.1	30.2

V. CONCLUSION AND FUTURE WORK

This study explored the application of deep learning techniques—specifically artificial neural networks (ANNs) and convolutional neural networks (CNNs)—for the classification of stream cipher keystreams generated by RC4, TRIVIUM, and ESPRESSO. Keystream samples were generated at multiple byte lengths (8, 24, 32, and 64 bytes), and models were trained on over 2^{17} samples per cipher type.

Despite the use of different model architectures and input sizes, all models consistently achieved around 30% accuracy, which aligns with random guessing in a 3-class classification problem. This result suggests that the statistical patterns in the keystreams are not easily distinguishable by standard deep learning approaches, highlighting the inherent strength of these ciphers against naive AI-based classification attacks.

Future work may involve exploring more advanced architectures such as recurrent neural networks (RNNs) or transformers, integrating feature engineering techniques, or analyzing additional cipher types to better understand the cryptographic properties that influence learnability.

The full codebase and dataset generation scripts used in this research are publicly available at:

<https://github.com/AAstik6/Streamcipher-Detection>

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